

Process Characterization Plan

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCP-003

Process Development / Manufacturing Science & Technology (MSAT)

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Table of contents

1 Purpose and scope	3
2 Related documents	3
3 Unit-operation description	4
4 Quality attributes in scope	5
5 Parameters and ranges to be characterized	6
6 Study design	6
7 Acceptance and decision criteria	7
8 Roles and responsibilities	7
9 Deliverables	8
10 Approvals	8
References	8

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1 Purpose and scope

This Process Characterization Plan (**PCP-003**) defines the Stage 1 (Process Design) characterization of the **A-Mab production bioreactor (Step 3)**, the fed-batch cell-culture operation that establishes the glycosylation, charge-variant and initial size-variant and process-impurity profile of the drug substance. The plan specifies the quality attributes in scope, the process parameters and ranges to be studied, the experimental designs, the analytical methods, and the acceptance and decision criteria by which each parameter will be classified and a design space established.

Scope. The 15,000-L commercial-scale production bioreactor, characterized on a qualified 2 L scale-down model (SDM). Seed-train expansion (Steps 1–2) and harvest (Step 4) are addressed only where they affect the production culture; their characterization is out of scope of this plan. This plan governs execution only — the results are reported in the paired **Process Characterization Report (PCR-003)**.

Basis. The unit-operation model, quality-attribute framework and risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009); the Stage 1 approach and report structure follow PDA Technical Report 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023); the Quality-by-Design framework follows ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012).

2 Related documents

This plan is executed under the master documents below and is paired with its report; the full package is cross-referenced for traceability.

Table 2: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-003	Process Characterization Report (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The governing procedures and validated analytical methods referenced by this plan are listed in Table 3 (numbers are placeholders in this synthetic set).

Table 3: Referenced standard operating procedures and analytical method validations.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

3 Unit-operation description

The production bioreactor is a fed-batch mammalian cell culture operated at 15,000 L working volume. It is the only upstream operation that forms product-quality CQAs and therefore the only upstream step used to define the upstream design space (CMC Biotech Working Group 2009); the seed expansion and harvest steps affect titer and process consistency but not product quality. Product quality established here — N-glycosylation (afucosylation, high mannose, galactosylation), charge variants, aggregate, and the host-cell protein (HCP) and DNA impurity load — is the input to the purification train (Figure 1).

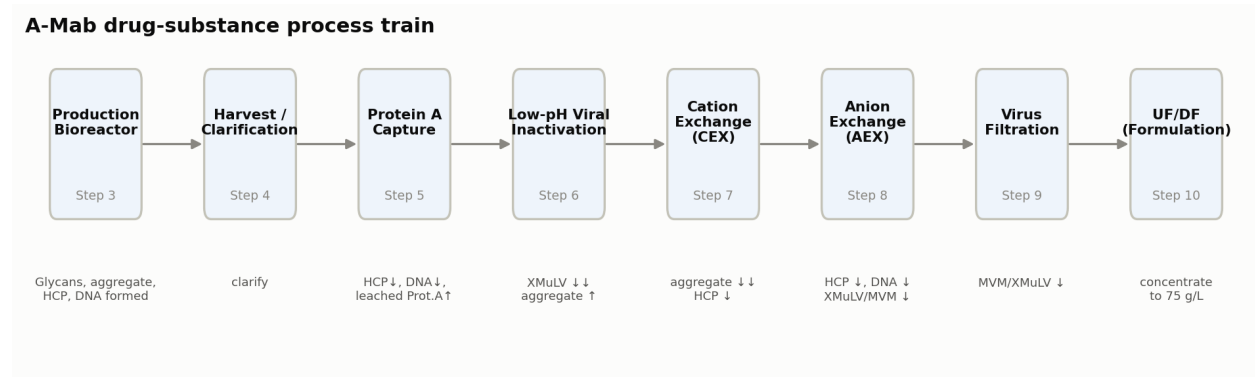


Figure 1: A-Mab drug-substance process flow (Steps 3–10); the production bioreactor (Step 3) forms the product-quality attributes characterized in this plan.

4 Quality attributes in scope

The characterization targets the CQAs that are set or principally controlled by the production bioreactor (Table 4). Criticality is assessed on the A-Mab continuum using Tool #1 (Risk Score = Impact × Uncertainty); attributes rated High (H) or Very High (VH) are treated as *Critical* (CMC Biotech Working Group 2009; International Council for Harmonisation 2023). Acceptance criteria are the drug-substance ranges carried from the CQA assessment and are the pass/fail basis for the study responses.

Table 4: CQAs in scope for the production-bioreactor characterization, with acceptance criteria and criticality (Tool #1 = Impact × Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

The study responses measured for these CQAs are afucosylation, galactosylation, high mannose and acidic charge variants (released assays), and aggregate (SEC); the harvest HCP and DNA impurity load is followed as a performance response since it is cleared downstream.

5 Parameters and ranges to be characterized

Nine process parameters are in scope (Table 5). Each is studied across the range shown (“Range studied”), which becomes the basis for the proven acceptable range (PAR); the normal operating range (NOR) is the tighter range at which the parameter will be controlled in routine manufacturing. The study type (multivariate vs univariate) is assigned by the risk-ranking-and-filtering screen of the Pre-Characterization Process Risk Assessment (**RA-001**): parameters with a credible main-effect and interaction risk to CQAs are studied multivariately; parameters expected to affect only performance are studied univariately or held at their set-point (CMC Biotech Working Group 2009).

Table 5: Production-bioreactor process parameters, set-points, characterization ranges and study type. Classification is an output of the study and is reported in PCR-003.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Culture pH	pH	6.85	6.6–7.1	6.75–6.95	multivariate
Culture temperature	°C	35	34–36	34.5–35.5	multivariate
Dissolved CO ₂ (pCO ₂)	mmHg	100	40–160	70–130	multivariate
Osmolality	mOsm	400	360–440	375–425	multivariate
Culture duration	days	17	15–19	16–18	multivariate
Dissolved oxygen	% sat	50	30–70	40–60	multivariate
Initial viable cell conc.	e6 cells/mL	1	0.5–1.5	0.8–1.2	univariate
Basal medium concentration	X	1.2	0.8–1.6	1–1.4	multivariate
Nutrient feed-1 volume	% WV	12	9–15	10–14	multivariate

Culture pH, temperature, dissolved CO₂, osmolality and culture duration are carried into the multivariate design as the parameters with the greatest prior-knowledge risk to the glycan and charge-variant CQAs. Dissolved oxygen, initial viable cell concentration and nutrient feed-1 volume are assessed for their effect on process performance (peak viable cell density, titer, viability); basal medium concentration is studied over a wide range as a general parameter.

6 Study design

Characterization uses a sequential design-of-experiments (DoE) strategy executed on the qualified 2 L SDM under **SOP-1002**, with the SDM qualified against at-scale reference data under **SOP-1001**:

1. **Screening design** — a resolution-V two-level fractional factorial in the five multivariate factors (culture pH, temperature, dissolved CO₂, osmolality and culture duration) with centre points, **19 runs total**, to identify the parameters and interactions with significant effects on the glycan, charge-variant and aggregate responses.
2. **Response-surface design** — a face-centred central-composite design in the dominant factors carried forward from screening (culture pH, temperature, culture duration and dissolved CO₂), **28 runs total**, to quantify main effects, two-factor interactions and curvature and to define the multivariate design space.

3. **Univariate assessment** — initial viable cell concentration is evaluated one factor at a time for its effect on process performance.

All runs are executed on the SDM at the commercial working volume equivalent and harvested on the nominal culture-duration criterion. Responses are measured by the validated methods in Table 3: released N-glycan map (afucosylation, high mannose, galactosylation) under **AMV-3010**, charge variants (acidic variants) under **AMV-3013**, aggregate (SEC) under **AMV-3011**, and the harvest HCP and residual-DNA load under **AMV-3012** and **AMV-3014**.

7 Acceptance and decision criteria

The study is acceptable when the fitted response-surface models are adequate (coefficient of determination and residual diagnostics per **SOP-1002**) and the design space satisfies every in-scope CQA simultaneously at the set-point and across the NOR. Each parameter is then classified per the A-Mab continuum (CMC Biotech Working Group 2009), consistent with the rule applied in the post-characterization risk assessment:

- A parameter whose variability significantly impacts a CQA is a **critical process parameter (CPP)**; where the impact is reliably controlled within the design space (low risk of leaving it) the parameter is a **well-controlled CPP (WC-CPP)**.
- A parameter that does not significantly impact a CQA but affects process performance or consistency is a **key process parameter (KPP)**; otherwise it is a **general process parameter (GPP)**.
- Consistent with the FMEA convention, a parameter that affects a CQA with Severity 8, or that carries an initial risk priority number > 72, is designated a CPP before mitigation by the control strategy.

The design space, parameter classification and the proven acceptable ranges established here provide the justification for the Stage 2 (PPQ) operating ranges and feed the overall control strategy.

8 Roles and responsibilities

Table 6: Roles and responsibilities for the production-bioreactor characterization.

Function	Responsibility
Process Development / MSAT	Author the plan; design, execute and analyze the DoE; author PCR-003
Analytical Development	Perform released and in-process testing under the referenced AMV methods
Manufacturing Science	Qualify the scale-down model; confirm commercial-scale representativeness
Quality Assurance	Approve the plan and the report; confirm parameter classifications

9 Deliverables

The deliverable of this plan is the **Process Characterization Report (PCR-003)**, which reports the executed designs, the significant effects, the design space, the parameter classification and the CQA outcomes. PCR-003 rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment.

10 Approvals

Table 7: Approval record (synthetic; signatures not applied).

Role	Name (synthetic)	Signature	Date
Author — MSAT	—	—	—
Reviewer — Analytical Development	—	—	—
Reviewer — Manufacturing Science	—	—	—
Approver — Quality Assurance	—	—	—

References

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